

Doing Bayesian Data Analysis

§4.1~4.2

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Classical Probability

- It is important to count the number of *all* possible outcomes (sample space).
- Then we count the number of outcomes which we are interested (event).
- Probability = $\#(\text{event})/\#(\text{sample space})$
- The same concept as shown in §4.2.1.2 Deriving a long-run relative frequency where each outcome is assumed to be equally likely.

17th Century

- Pascal and Fermat's correspondence (1654 July)
- De Mere: "At least on 6 in 4 rolls" vs "at least on double-six in 24 throws of 2 dice" (crucial)

Truth is the same of Toulouse and at Paris. --by Pascal

Why Incident in Python

- Check an example:

```
i=1;  
for x in range(5):  
    print(x)  
i=i+1
```

What will `i` be in the end?

Long-Run Relative Frequency

- In April 1940, while visiting in-laws in Copenhagen, Kerrich was caught up in the Nazi invasion and interned in Hald Ege, Viborg, Midtjylland.
- Kerrich tosses 10,000 times to demonstrate Jacob Bernoulli's famous law of large numbers.
- The proportion of heads obtained asymptotically approached the theoretical value of 50 percent (the precise number obtained was 5,067).

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Random Number Generator

- The Python programming language offers pseudo-random number generators (PRNG) to mimic the properties of random processes.
- The PRNGs are in fact deterministic: This is why `np.random.seed()` ensures that the sequence of random numbers generated afterward is the same.
- `np.random.randint(0, __)` randomly generates a integer between 0 to 1. ← What is the number we place in the second argument?

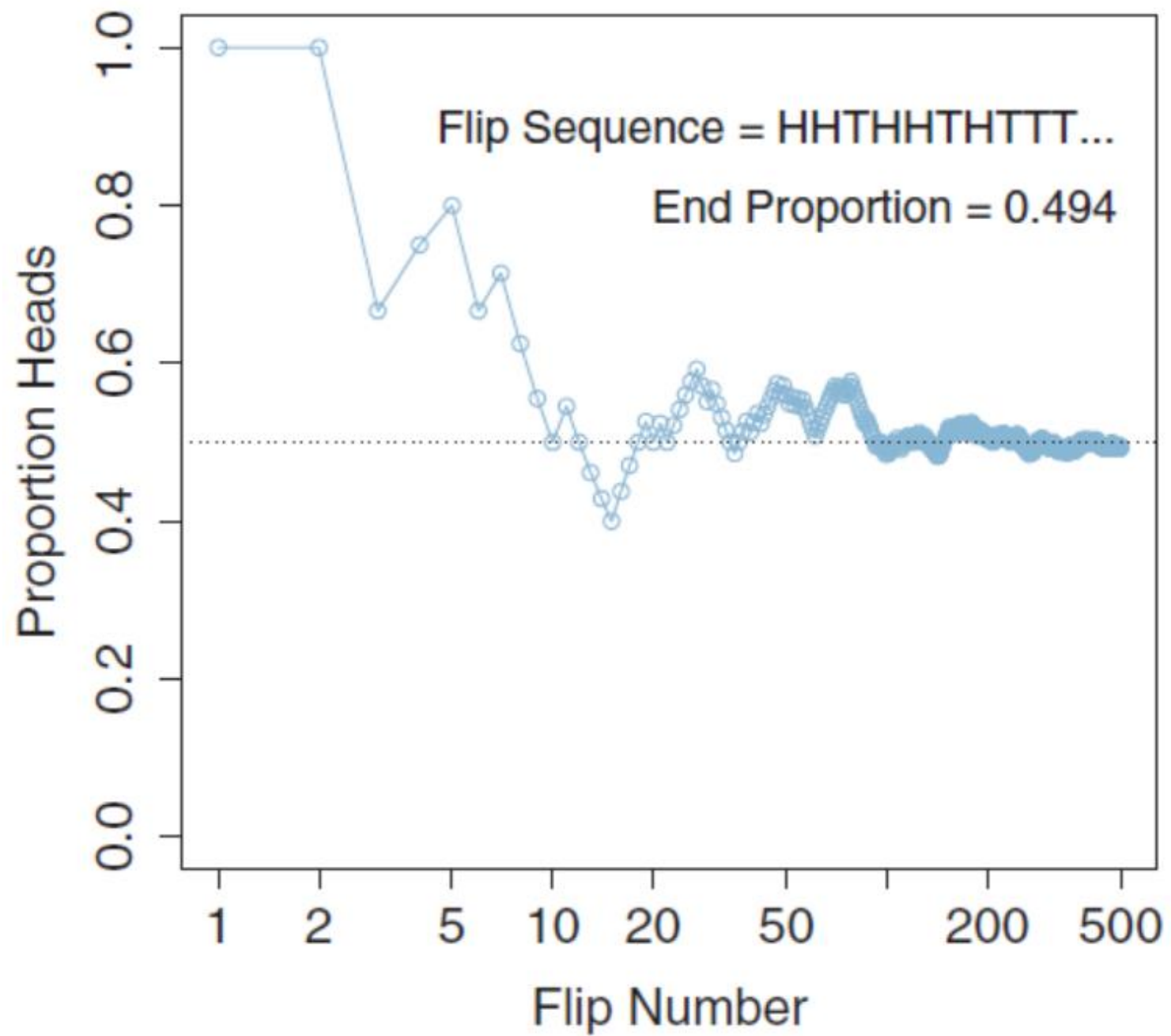
Assessing Probability

2. Relative-Frequency Probability (客觀的機率理論)

長期重複試行的隨機實驗中，事件的相對次數會漸趨穩定。而此一次數的比率即為該事件的機率。

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

即資料越多越接近真實機率

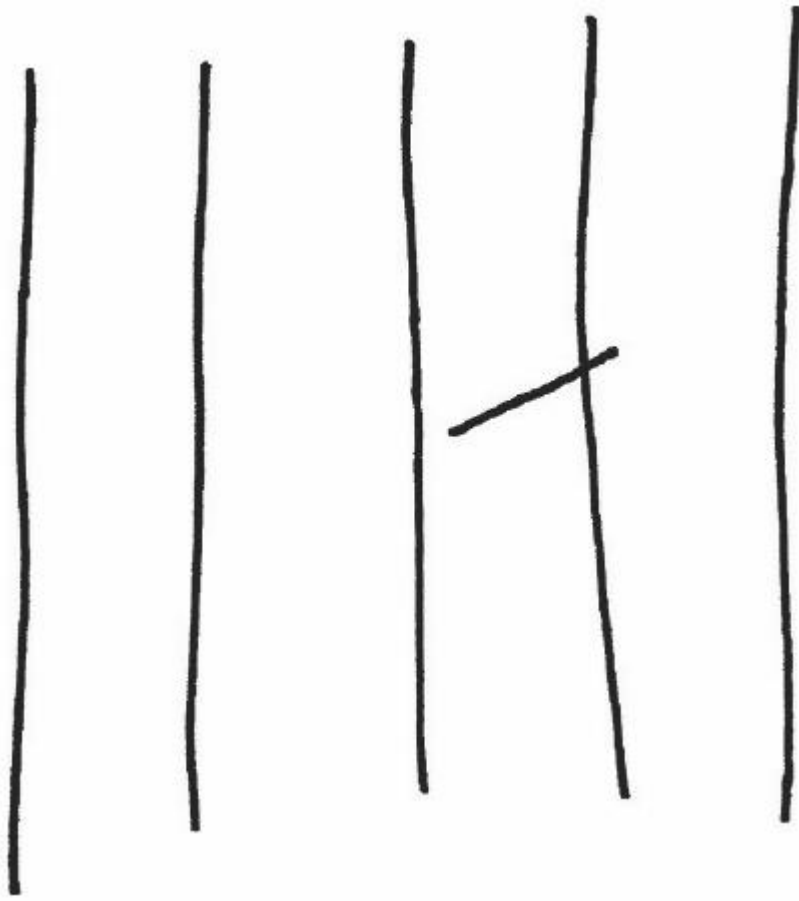


Caveats

- At the end of the long sequence, the proportion of heads is near 0.5 but not necessarily exactly equal to 0.5.
- There is no guarantee that the relative frequency of an event will match the true underlying probability of the event.
- That's why we say we are *approximating* the probability by the long-run relative frequency.  Notice that approximation is not learning/inference.

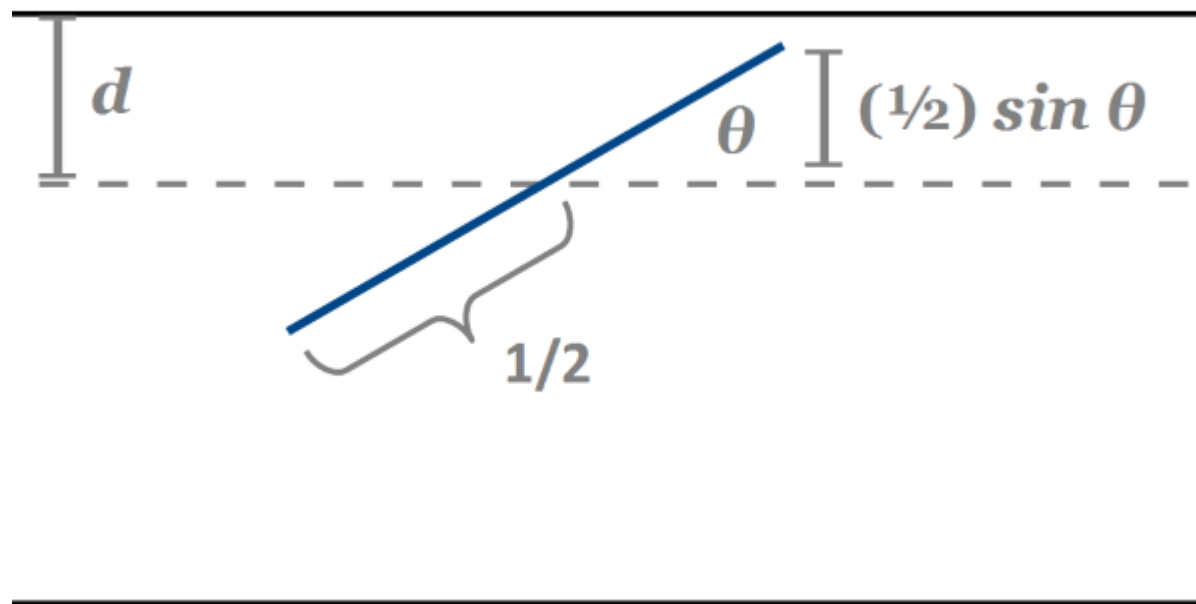
Buffon's Needle Problem

- Georges-Louis Leclerc, Comte de Buffon (1777)



Buffon's Needle Problem

- The needle will hit the line if the closest distance to a line (D) is less than or equal to $1/2$ times the sine of θ . That is, $D \leq (1/2)\sin(\theta)$.



Assessing Probability

3. Subjective probability

An individual opinion or degree of belief about the probability of occurrence

EX. 對某市長候選人當選的可能性, 明天某股票股價會上漲的機率等。

Probability Axioms

1. If A is any event in the sample space S , then for each $E_i \in S$,

$$0 \leq P(E_i) \leq 1$$

2. Let E_i be the exclusive events in S . Then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

3. $P(S) = 1$

Probability Rules

- The **Complement rule** (餘集合定理):

$$P(\bar{A}) = 1 - P(A) \quad \text{i.e., } P(A) + P(\bar{A}) = 1$$

Ex. De Mere's problem (1717)

- The **Addition rule** (加法定理):
 - The probability of the union of two events is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Extract Subjective Probability

- Consider the following two gambles that you can choose from:
Gamble A: You get \$100 if there is a traffic stopping snowstorm in Indianapolis next New Year's Day.
Gamble B: You get \$100 if you draw a red marble from a sack of marbles with 5 red and 5 white marbles.
- If you prefer Gamble B, we know that your subjective belief about the probability of traffic-stopping snowstorm is less than 0.5.

Extract Subjective Probability

- Consider the following two gambles that you can choose from:
Gamble A: You get \$100 if there is a traffic stopping snowstorm in Indianapolis next New Year's Day.
Gamble B: You get \$100 if you draw a red marble from a sack of marbles with 1 red and 9 white marbles.
- If you prefer Gamble A, we know that your subjective belief about the probability of traffic-stopping snowstorm is between 0.1 and 0.5.

Monty Hall Problem Revisited

- Ex. Monty Hall problem (3-door problem)
Let's Make a Deal



Monty Hall Problem Revisited

- Ex. Monty Hall problem (3-door problem)



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For the Existence of God

- In David Hume's publication *Of Miracles* (1748):

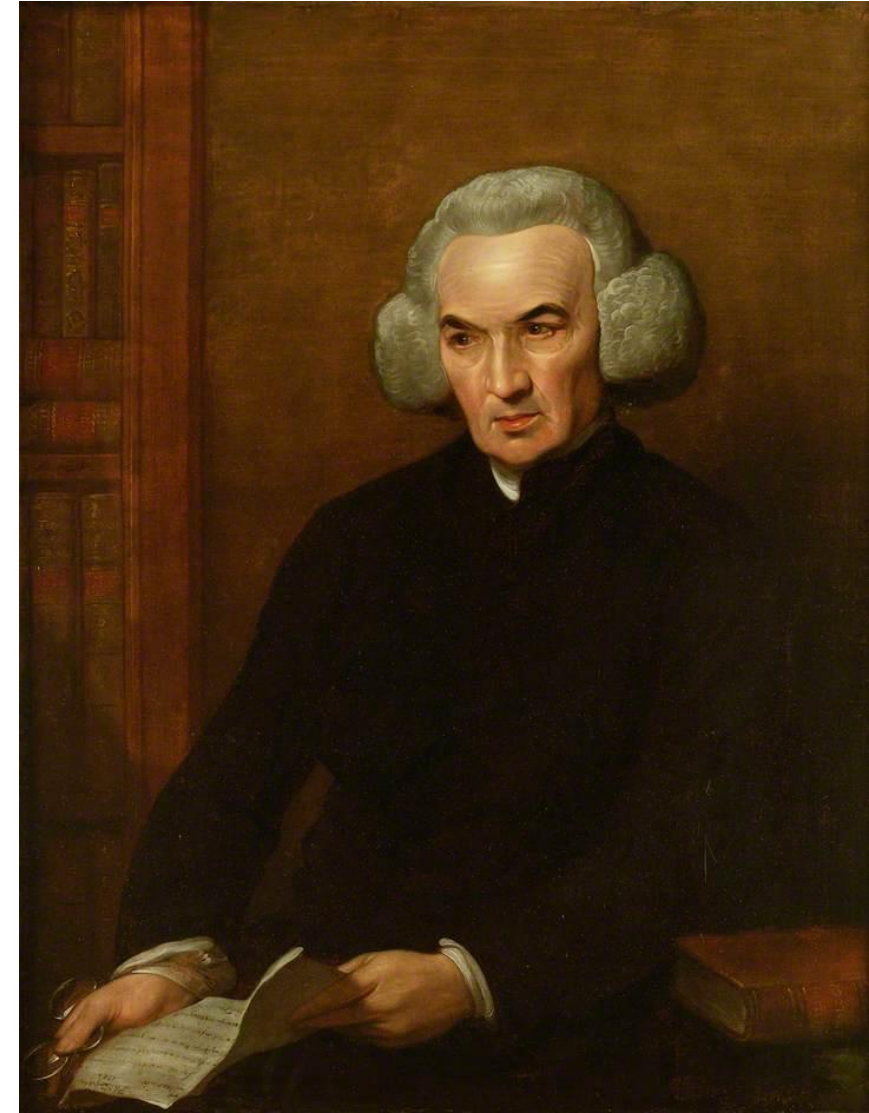
“No testimony is sufficient to establish a miracle, unless the testimony be of such a kind, that its falsehood would be more miraculous than the fact which it endeavors to establish.”



For the Existence of God

- Richard Price (1767):

Based on a million observations, there is a 50% chance the true probability of the tide not coming is between $1/600,000$ and $1/3,000,000$ —using Bayes' rule.



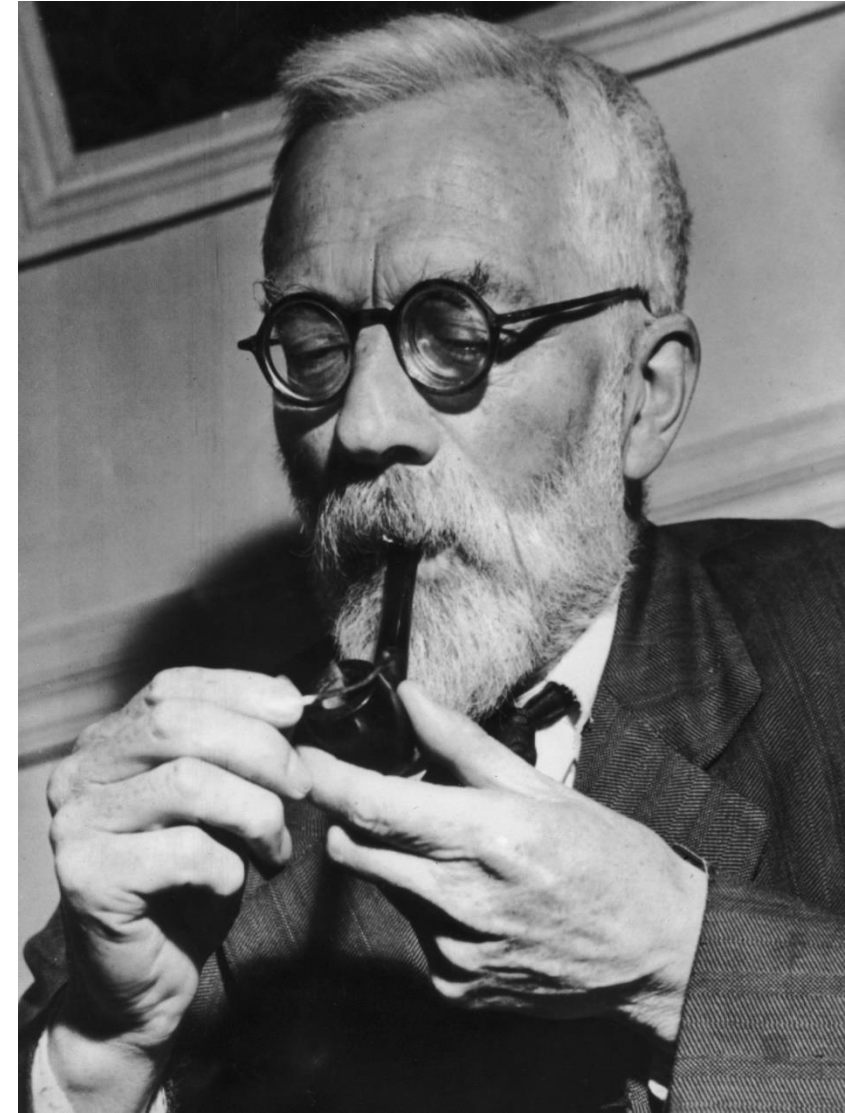
History of Bayes' Rule

- Thomas Bayes (1702-1761) was a mathematician and Presbyterian minister in England.
- Bayes' rule was published in 1763 Price reads 'An Essay towards solving a Problem in the Doctrine of Chances' to the Royal Society on December 23rd, 1763.



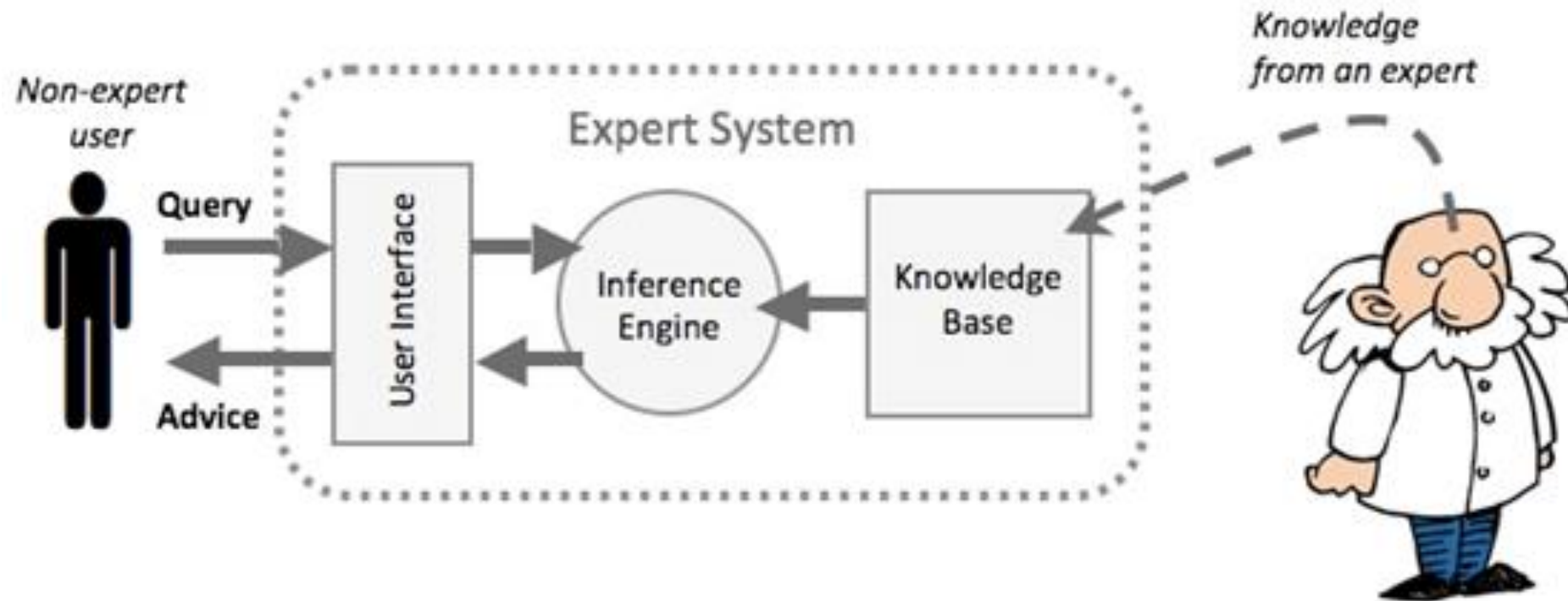
Frequentist

- Frequentist, identified with Ronald Fisher (1890-1962), does not use Bayes' rule for inference and decisions.
- In 1922, Fisher introduced the method of maximum likelihood to replace the inverse probability.



Revival of Bayesian

- The dominant Fisherian approach of the 20th century is giving way in the 21st century to a Bayesian approach that had its genesis in the 18th century.



Reallocation of Credibility

- On a typical day at your location, what is the probability that it is cloudy?
- Suppose you are told it is raining, now what is the probability that it is cloudy?
- We can be pretty sure that
$$p(\text{cloudy}) < p(\text{cloudy}|\text{raining}).$$

Human Reasoning

- Suppose instead you are told that everyone outside is wearing sunglasses, what is the probability that it is cloudy?
- Most likely, it is true that
$$p(\text{cloudy}) > p(\text{cloudy}|\text{sunglass}).$$
- Notice how we human beings have reasoned in this example.

Bayes' Rule

- Bayes' rule is merely the mathematical relation between the prior allocation of credibility and the posterior reallocation of credibility conditional on data.

- From the definition of conditional probability,

$$p(c|r) = \frac{p(r, c)}{p(r)}$$

i.e., the probability that r and c happen together relative to the probability that r happens at all.



Only a condition

Derivation

- Notice that by definition we also have

$$p(r|c) = \frac{p(r, c)}{p(c)} \quad \leftarrow \text{Marginal probability}$$

- Multiplying both sides by $p(r)$ and $p(c)$, respectively, gives

$$\begin{aligned} p(c|r)p(r) &= p(r, c) = p(r|c)p(c) \\ \Rightarrow p(c|r) &= \frac{p(r|c)p(c)}{p(r)} \end{aligned}$$

which is called *Bayes' rule*.

Bayes' Rule

- We can further re-write the denominator in terms of $p(r|c)$:

$$p(c|r) = \frac{p(r|c)p(c)}{\sum_{c^*} p(r|c^*)p(c^*)}$$

- The c in the numerator is a specific fixed value, whereas the c^* in the denominator is a variable that takes on all possible values.
- $p(c|r)$ is called posterior; $p(c)$ is prior; $p(r|c)$ is likelihood; $p(r)$ is marginal likelihood (evidence).

Intuition of Bayes Rule

- Without knowing anything about a person's eye color, all we believe about hair colors is expressed by the marginal probabilities at the bottom.

Eye color	Hair color				Marginal (Eye color)
	Black	Brunette	Red	Blond	
Brown	0.11	0.20	0.04	0.01	0.37
Blue	0.03	0.14	0.03	0.16	0.36
Hazel	0.03	0.09	0.02	0.02	0.16
Green	0.01	0.05	0.02	0.03	0.11
Marginal (hair color)	0.18	0.48	0.12	0.21	1.0

Intuition of Bayes Rule

- If we are told that the selected person's eyes are blue, we can focus our attention on the Blue row.
- Notice that we have gone from the “prior” (marginal) beliefs about hair color before knowing eye color, to the “posterior” (conditional) beliefs.

Eye color	Hair color				Marginal (Eye color)
	Black	Brunette	Red	Blond	
Blue	$0.03/0.36$ $= 0.08$	$0.14/0.36$ $= 0.39$	$0.03/0.36$ $= 0.08$	$0.16/0.36$ $= 0.45$	$0.36/0.36 = 1.0$

Quiz 2b

- Please complete the code for a four-door problem to calculate the number of winning a car if you always switch to another randomly chosen door.
- What is your guess for the true probability of winning a car when you always choose to switch?
- Due by 12:00 PM on 9 March, 2026:
<https://forms.gle/v33vWb8poyg3M6abA>